

Program 1

Use the **keras** package.

Utilise the Long Short-Term Memory units to train a neural network capable of predicting sequential data. Predictions are of form: "given this historical sequence of values, what is the value for the next day?"

Choose a dataset with observations over a long period of time. Data from the stock market, cryptocurrencies, temperature readings, or similar is recommended. Make sure that the values fluctuate (that is, not just continuously going up/down with some noise), and look reasonably unpredictable (up to your judgement). Present and cite the data that you are using.

For example: [this synthetic stock market dataset](#), or [this ORLEN stock price dataset](#).

Split the dataset into training and testing sets. Note that the split should not be random per-value as in other projects, but rather splitting the data into two sequences "before X" and "after X", with the first becoming the training set and the latter - the testing set. See Figure 1.

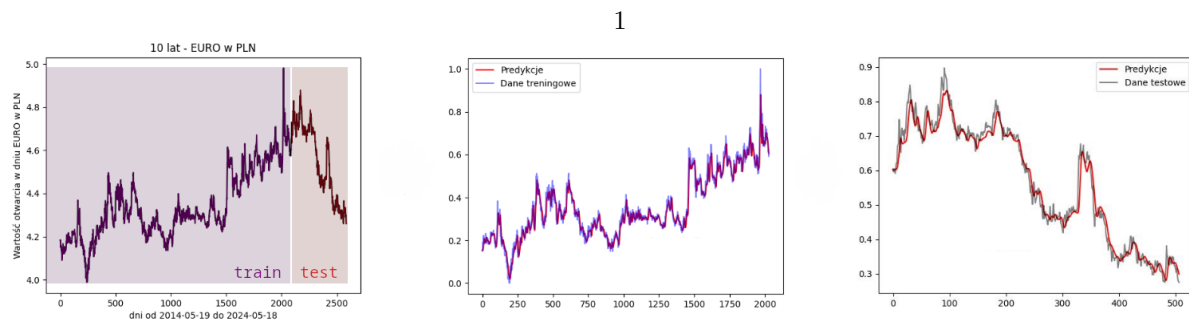


Figure 1: Sequential data split

Design and fit the model. See the [Keras documentation for the LSTM layer](#).

Evaluate the model. Plot the model predictions over both sets of data. Plot and note the final loss during training.

Grading

- 8 points - Network architecture, training, and testing¹
- 2 points - Graphs and analysis

10 points total.

Work to be submitted

All source code files (.py, .ipynb),

A .pdf document with the visualisations and written analysis,

The exported model file.

If the submission does not include all of the above, it will not be checked.

Do not send archived files, IDE configuration, cache, etc.

¹The grade for this rubric is directly proportional to the quality of predictions.